Multivariable Calculus formula sheet

The formulas of Multivariable Calculus (Mathematical Analysis 2) with their hypotheses: limits and derivatives in several variables, the implicit function theorem, Lagrange multipliers, line, double, triple and surface integrals, Green, Gauss and Stokes, power series and Fourier series.

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Limits and continuity in two variables

The proofs and the examples are in the chapter Functions of several variables.

Norm, distance and neighborhood

$$\|(x,y)\| = \sqrt{x^2+y^2}, \qquad d(P,Q) = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$$

$$B(P_0, r) = \{P : \|P - P_0\| < r\}$$

The formulas hold for $P = (x_1, y_1)$, $Q = (x_2, y_2)$ and $r > 0$. In $\mathbb{R}^n$ the norm is the square root of the sum of the squares of the n components.

Conditions for the natural domain

Expression	Condition
$\dfrac{A}{B}$	$B \neq 0$
$\sqrt[2k]{A}$	$A \geq 0$
$\ln A$	$A > 0$
$\arcsin A$, $\arccos A$	$-1 \leq A \leq 1$

The conditions present in the expression of $f(x,y)$ must be satisfied together, therefore they are collected in a system.

Restriction theorem

$$\lim_{(x,y)\to P_0} f(x,y) = L \implies \lim_{t\to t_0} f(\gamma(t)) = L$$

The implication holds for every curve γ in the domain with $\gamma(t) \to P_0$ and $\gamma(t) \neq P_0$ as $t \to t_0$. Therefore, if two restrictions have different limits, the limit of f does not exist. At the origin the most used restrictions are $f(x, mx)$, $f(0, y)$ and $f(x, x^2)$.

Squeeze theorem

$$|f(x,y)-L| \le g(x,y), \qquad \lim_{(x,y)\to P_0} g(x,y)=0 \implies \lim_{(x,y)\to P_0} f(x,y)=L$$

The inequality must hold in a neighborhood of P_0, except at most P_0 itself. The theorem is used to prove that a limit exists.

Useful bounds

$$|x| \le \sqrt{x^2+y^2}, \qquad x^2 \le x^2+y^2, \qquad |xy| \le \frac{x^2+y^2}{2}$$

$$0 \le \frac{x^2}{x^2+y^2} \le 1, \qquad |\sin t| \le |t|, \qquad |\sin t| \le 1$$

They hold for every (x,y) and every real t, the quotient for $(x,y) \neq (0,0)$, and they also hold with y in place of x.

Bounded times infinitesimal

$$|f(x,y)| \le C, \qquad \lim_{(x,y)\to P_0} g(x,y)=0 \implies \lim_{(x,y)\to P_0} f(x,y)\,g(x,y)=0$$

The bound $|f| \le C$ must hold in a neighborhood of P_0. The sine and cosine of any argument are bounded, and so are the quotients $\dfrac{x^2}{x^2+y^2}$ and $\dfrac{xy}{x^2+y^2}$.

Limits in polar coordinates

$$x = x_0+\rho\cos\theta, \qquad y = y_0+\rho\sin\theta, \qquad \rho>0,\ \theta\in[0,2\pi)$$

$$\begin{cases} |f(x_0+\rho\cos\theta,\, y_0+\rho\sin\theta)-L| \le g(\rho) \ \text{ for every } \theta \\ g(\rho)\to 0 \ \text{ as } \rho\to 0^+ \end{cases} \implies \lim_{(x,y)\to P_0} f(x,y)=L$$

The function $g(\rho)$ must be **independent of** θ. With $P_0=(0,0)$ one has $x^2+y^2=\rho^2$.

Standard limits with an infinitesimal argument

$$\lim_{(x,y)\to P_0} \frac{\sin g(x,y)}{g(x,y)} = 1, \qquad \lim_{(x,y)\to P_0} \frac{1-\cos g(x,y)}{g(x,y)^2} = \frac{1}{2}$$

$$\lim_{(x,y)\to P_0} \frac{\ln\big(1+g(x,y)\big)}{g(x,y)} = 1, \qquad \lim_{(x,y)\to P_0} \frac{e^{g(x,y)}-1}{g(x,y)} = 1$$

They hold if $g(x,y)\to 0$ as $(x,y)\to P_0$ and $g(x,y)\neq 0$ in a neighborhood of P_0 with P_0 removed. At the origin the typical argument is $g=x^2+y^2$.

Continuity and continuous extension

$$f \text{ continuous at } P_0 \iff \lim_{(x,y)\to P_0} f(x,y) = f(P_0)$$

The definition holds for P_0 in the domain and an accumulation point of the domain. Polynomials and elementary functions are continuous on their domain, and so are sums, products, quotients with nonzero denominator and compositions of continuous functions. If P_0 does not belong to the domain and the limit exists, is finite and equals L, the function is extended by continuity by setting $\tilde{f}(P_0) = L$.

Warning

The restrictions along straight lines can all have the same limit even when the limit does not exist. For $f(x,y) = \dfrac{x^2 y}{x^4 + y^2}$ at the origin one has $f(x, mx) = \dfrac{mx}{x^2 + m^2} \to 0$ for every m and $f(0, y) = 0$, while $f(x, x^2) = \dfrac{1}{2}$ for every $x \neq 0$. Therefore the limit does not exist.

Partial derivatives, differentiability and tangent plane

The complete statements and the proofs are in the chapter Differential calculus in several variables.

Partial derivatives

$$f_x(x_0, y_0) = \lim_{t\to 0} \frac{f(x_0 + t,\, y_0) - f(x_0, y_0)}{t}$$

$$f_y(x_0, y_0) = \lim_{t\to 0} \frac{f(x_0,\, y_0 + t) - f(x_0, y_0)}{t}$$

The derivatives exist if the limits exist and are finite, with (x_0, y_0) interior to the domain. In the computation of f_x the variable y is treated as a constant. At the junction points of a piecewise-defined function the definition is used.

Gradient

$$\nabla f(x_0, y_0) = \big(f_x(x_0, y_0),\ f_y(x_0, y_0)\big), \qquad \nabla f = (f_x,\ f_y,\ f_z) \ \text{ in } \mathbb{R}^3$$

The gradient is defined at the points where f has partial derivatives.

Differentiability

$$\lim_{(h,k)\to(0,0)} \frac{f(x_0+h,\,y_0+k)-f(x_0,y_0)-f_x(x_0,y_0)\,h-f_y(x_0,y_0)\,k}{\sqrt{h^2+k^2}}=0$$

$$f(x_0+h,\,y_0+k)=f(x_0,y_0)+f_x\,h+f_y\,k+o\big(\sqrt{h^2+k^2}\big)$$

The two formulations are equivalent and require that f have partial derivatives at (x_0,y_0). The differential is the linear map $(h,k)\mapsto f_x(x_0,y_0)\,h+f_y(x_0,y_0)\,k$.

Total differential theorem

$$f_x,\ f_y \text{ continuous at } (x_0,y_0) \implies f \text{ differentiable at } (x_0,y_0)$$

The partial derivatives must exist in a neighborhood of the point. In particular a function of class $C^1(A)$ is differentiable at every point of the open set A.

Scheme of implications

$$C^1 \implies \text{differentiable} \implies \text{continuous}, \qquad \text{differentiable} \implies \text{has partial derivatives}$$

None of the implications can be reversed. A function discontinuous at a point is therefore not differentiable at that point.

Tangent plane to the graph

$$z=f(x_0,y_0)+f_x(x_0,y_0)\,(x-x_0)+f_y(x_0,y_0)\,(y-y_0)$$

The equation represents the tangent plane to the graph at the point $\big(x_0,y_0,f(x_0,y_0)\big)$ if f is differentiable at (x_0,y_0). A normal vector to the plane is $\big(f_x(x_0,y_0),\ f_y(x_0,y_0),\ -1\big)$.

Directional derivative, chain rule and Taylor's formula

The formulas are proved in the chapter Differential calculus in several variables.

Directional derivative

$$D_{\mathbf{v}}f(x_0,y_0)=\lim_{t\to0}\frac{f(x_0+tv_1,\,y_0+tv_2)-f(x_0,y_0)}{t}$$

The vector $\mathbf{v}=(v_1,v_2)$ is a unit vector, that is, $v_1^2+v_2^2=1$. If the direction is assigned by a vector $\mathbf{w}$ of arbitrary length, one uses $\mathbf{v}=\dfrac{\mathbf{w}}{|\mathbf{w}|}$.

Gradient formula

$$D_{\mathbf{v}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{v} = f_x(x_0, y_0)\, v_1 + f_y(x_0, y_0)\, v_2$$

The formula holds if f is **differentiable** at (x_0, y_0). At the other points the directional derivative is computed with the definition.

Direction of steepest ascent

$$\max_{|\mathbf{v}|=1} D_{\mathbf{v}} f(x_0, y_0) = |\nabla f(x_0, y_0)|, \qquad \mathbf{v} = \frac{\nabla f(x_0, y_0)}{|\nabla f(x_0, y_0)|}$$

It holds for f differentiable with $\nabla f(x_0, y_0) \neq (0, 0)$. The minimum value $-|\nabla f(x_0, y_0)|$ is attained along the opposite unit vector; along the directions orthogonal to the gradient the directional derivative is zero, hence the gradient is orthogonal to the level curve through the point.

Chain rule with one parameter

$$\frac{d}{dt} f\big(x(t), y(t)\big) = f_x\big(x(t), y(t)\big)\, x'(t) + f_y\big(x(t), y(t)\big)\, y'(t)$$

It holds if $x(t)$ and $y(t)$ are differentiable at t and f is differentiable at the point $\big(x(t), y(t)\big)$. In vector form the derivative is $\nabla f\big(x(t), y(t)\big) \cdot \big(x'(t), y'(t)\big)$.

Chain rule with two parameters

$$\frac{\partial F}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s}, \qquad \frac{\partial F}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}$$

They hold for $F(s, t) = f\big(x(s, t), y(s, t)\big)$, with f differentiable and x, y having partial derivatives; the derivatives of f are evaluated at $\big(x(s, t), y(s, t)\big)$.

Second derivatives and Schwarz's theorem

$$f_{xy} = \frac{\partial}{\partial y} f_x, \qquad f_{yx} = \frac{\partial}{\partial x} f_y, \qquad f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$$

The equality of the mixed derivatives holds if f_{xy} and f_{yx} exist in a neighborhood of (x_0, y_0) and are continuous at the point, in particular for every f of class C^2.

Hessian matrix

$$H_f(x_0, y_0) = \begin{pmatrix} f_{xx}(x_0, y_0) & f_{xy}(x_0, y_0) \\ f_{yx}(x_0, y_0) & f_{yy}(x_0, y_0) \end{pmatrix}, \qquad \det H_f = f_{xx} f_{yy} - f_{xy}^2$$

For f of class C^2 the matrix is symmetric, by Schwarz's theorem, and the formula for the determinant uses $f_{xy} = f_{yx}$.

Second-order Taylor formula

$$f(x_0+h,\,y_0+k) = f(x_0,y_0) + f_x(x_0,y_0)\,h + f_y(x_0,y_0)\,k + \frac{1}{2}\Big(f_{xx}(x_0,y_0)\,h^2 + 2f_{xy}(x_0,y_0)\,hk + f_{yy}(x_0,y_0)\,k^2\Big) + o\big(h^2+k^2\big)$$

It holds for f of class C^2 in a neighborhood of (x_0,y_0) and for $(h,k)\to(0,0)$, with $h = x - x_0$ and $k = y - y_0$. The second-degree term is $\frac{1}{2}\,(h,k)\,H_f(x_0,y_0)\,(h,k)^T$.

Implicit functions and the implicit function theorem

The proofs and the examples are in the chapter Implicit functions.

Implicit function theorem in two variables

$$\begin{cases} F(x_0,y_0) = 0 \\ F_y(x_0,y_0) \neq 0 \end{cases} \implies \varphi'(x) = -\frac{F_x(x,\varphi(x))}{F_y(x,\varphi(x))}$$

It holds for F of class C^1 on an open set containing (x_0,y_0). Under the hypotheses there exist a neighborhood I of x_0, a neighborhood J of y_0 and a unique function $\varphi : I \to J$ of class C^1 with $\varphi(x_0) = y_0$ and $F(x,\varphi(x)) = 0$ for every $x \in I$; in particular $\varphi'(x_0) = -\dfrac{F_x(x_0,y_0)}{F_y(x_0,y_0)}$.

Implicit function of the variable y

$$\begin{cases} F(x_0,y_0) = 0 \\ F_x(x_0,y_0) \neq 0 \end{cases} \implies \psi'(y) = -\frac{F_y(\psi(y),y)}{F_x(\psi(y),y)}$$

In this case near the point the curve $F = 0$ is the graph of $x = \psi(y)$. At the points where F_x and F_y are both zero the theorem does not apply in either form, and the points are called singular.

Second derivative of the implicit function

$$\varphi''(x) = -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3}$$

It holds for F of class C^2 under the hypotheses of the implicit function theorem, with all the partial derivatives evaluated at $(x,\varphi(x))$. The same result is obtained by differentiating twice the identity $F(x,\varphi(x)) = 0$.

Stationary points of the implicit function

$$\varphi'(x_0) = 0 \iff F_x(x_0, y_0) = 0, \qquad \varphi''(x_0) = -\frac{F_{xx}(x_0, y_0)}{F_y(x_0, y_0)}$$

They hold for F of class C^2, $F(x_0, y_0) = 0$ and $F_y(x_0, y_0) \neq 0$; the second formula holds at the points with $F_x(x_0, y_0) = 0$. If $\varphi''(x_0) < 0$, the point x_0 is a local maximum point for φ; if $\varphi''(x_0) > 0$, it is a local minimum point.

Tangent line to a level curve

$$F_x(x_0, y_0)\,(x - x_0) + F_y(x_0, y_0)\,(y - y_0) = 0$$

It holds for F of class C^1 with $F(x_0, y_0) = 0$ and $\nabla F(x_0, y_0) \neq (0, 0)$. The gradient $\nabla F(x_0, y_0)$ is orthogonal to the curve; if $F_y(x_0, y_0) = 0$, the line is vertical.

Implicit function theorem in three variables

$$\begin{cases} F(x_0, y_0, z_0) = 0 \\ F_z(x_0, y_0, z_0) \neq 0 \end{cases} \implies g_x = -\frac{F_x}{F_z}, \qquad g_y = -\frac{F_y}{F_z}$$

It holds for F of class C^1. Near (x_0, y_0, z_0) the surface $F = 0$ is the graph of a function $z = g(x, y)$ of class C^1 with $g(x_0, y_0) = z_0$, and the derivatives of F are evaluated at $(x, y, g(x, y))$.

Tangent plane to a level surface

$$F_x(P_0)\,(x - x_0) + F_y(P_0)\,(y - y_0) + F_z(P_0)\,(z - z_0) = 0$$

It holds for F of class C^1 with $F(P_0) = 0$ and $\nabla F(P_0) \neq (0, 0, 0)$, where $P_0 = (x_0, y_0, z_0)$. The gradient $\nabla F(P_0)$ is the normal vector to the surface.

Implicit function theorem for systems

$$\frac{\partial(F, G)}{\partial(y, z)}(P_0) = \det \begin{pmatrix} F_y & F_z \\ G_y & G_z \end{pmatrix}(P_0) \neq 0$$

$$\begin{cases} F_x + F_y\, y'(x) + F_z\, z'(x) = 0 \\ G_x + G_y\, y'(x) + G_z\, z'(x) = 0 \end{cases}$$

If F and G are of class C^1, P_0 solves the system $F = 0$, $G = 0$ and the Jacobian determinant is nonzero, near P_0 the system defines $y = y(x)$ and $z = z(x)$ of class C^1. The derivatives $y'(x_0)$ and $z'(x_0)$ solve the linear system in the unknowns y' and z', with the derivatives of F and G evaluated at P_0.

Unconstrained and constrained maxima and minima

The methods and the complete examples are in the chapter Unconstrained and constrained maxima and minima.

Critical points and Fermat's theorem

$$\begin{cases} f_x(x_0, y_0) = 0 \\ f_y(x_0, y_0) = 0 \end{cases}$$

If (x_0, y_0) is a local extremum point interior to the domain and f has partial derivatives there, the system is satisfied. The unconstrained extrema are therefore sought among the critical points and the interior points at which f has no partial derivatives.

Classification with the Hessian matrix

$$\begin{aligned} \det H > 0,\ f_{xx} > 0 &\implies \text{local minimum} \\ \det H > 0,\ f_{xx} < 0 &\implies \text{local maximum} \\ \det H < 0 &\implies \text{saddle point} \end{aligned}$$

It holds for f of class C^2 at a critical point, with $H = H_f(x_0, y_0)$ and the derivatives evaluated at the point. If $\det H = 0$, the test is inconclusive.

Degenerate Hessian: study of the sign

$$\begin{aligned} f(x,y) - f(x_0,y_0) \geq 0 \ \text{ near } (x_0,y_0) &\implies \text{local minimum} \\ f(x,y) - f(x_0,y_0) \leq 0 \ \text{ near } (x_0,y_0) &\implies \text{local maximum} \end{aligned}$$

The direct study is used when $\det H = 0$. If in *every* neighborhood the difference takes both signs, the point is a saddle point; to prove it, two curves through the point along which the difference has opposite signs suffice.

Weierstrass's theorem and absolute extrema

$$K \text{ compact},\ f \text{ continuous on } K \implies f \text{ attains an absolute maximum and an absolute minimum}$$

A subset of $\mathbb{R}^n$ is compact if it is closed and bounded. The candidates are the interior critical points, the interior points at which f has no partial derivatives and the extremum points of the restriction to the boundary, vertices included; the comparison of the values of f at the candidates gives the maximum and the minimum.

Restriction to the boundary

$$g(t) = f(x_0 + r\cos t,\ y_0 + r\sin t), \qquad t \in [0, 2\pi]$$

It is the restriction of f to the circle with center (x_0, y_0) and radius r. On the segment $y = c$ with $x \in [a, b]$ the restriction is $g(x) = f(x, c)$, to be studied on $[a, b]$ with the endpoints of the interval among the candidates.

Lagrange multipliers with one constraint

$$\begin{cases} f_x(x,y) = \lambda\, g_x(x,y) \\ f_y(x,y) = \lambda\, g_y(x,y) \\ g(x,y) = 0 \end{cases}$$

It holds for f and g of class C^1. If (x_0, y_0) is a constrained extremum point with $\nabla g(x_0, y_0) \neq (0, 0)$, there exists λ such that (x_0, y_0, λ) solves the system. If the constraint is compact, the maximum and the minimum are obtained by comparing the values of f at the solutions and at the points of the constraint with $\nabla g = (0, 0)$.

Lagrange multipliers with two constraints

$$\begin{cases} \nabla f(x,y,z) = \lambda\, \nabla g(x,y,z) + \mu\, \nabla h(x,y,z) \\ g(x,y,z) = 0 \\ h(x,y,z) = 0 \end{cases}$$

It holds for f, g, h of class C^1 at the points where ∇g and ∇h are linearly independent. The system has five equations in the unknowns x, y, z, λ, μ.

Distance from a point

$$d^2 = (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2$$

The square root is increasing, therefore the distance and its square have the same maximum and minimum points. In minimum distance problems one therefore optimizes d^2, which does not contain the root.

Warning

The points of the constraint where $\nabla g = (0, 0)$ do not appear among the solutions of the Lagrange system. With the constraint $g(x, y) = y^3 - x^2 = 0$ the function $f(x, y) = y$ has a constrained minimum at the origin, because on the constraint $y = x^{\frac{2}{3}} \geq 0$. At the origin the system requires $(0, 1) = \lambda\,(0, 0)$, which has no solution; the point must be added to the candidates.

Curves, length and line integrals

The definitions and the proofs are in the chapter Curves and line integrals.

Regular curve and unit tangent vector

$$\gamma'(t) = \big(x'(t),\, y'(t),\, z'(t)\big) \neq \mathbf{0}, \qquad T(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$$

The curve $\gamma : [a,b] \to \mathbb{R}^n$ is regular if it is of class C^1 and $\gamma'(t) \neq \mathbf{0}$ for every $t \in (a,b)$. It is piecewise regular if it is continuous and regular on each interval of a partition of $[a,b]$.

Tangent line to a parametric curve

$$r(s) = \gamma(t_0) + s\,\gamma'(t_0), \qquad s \in \mathbb{R}$$

It holds at the points with $\gamma'(t_0) \neq \mathbf{0}$. For a plane curve with $x'(t_0) \neq 0$ the slope of the tangent line is $\dfrac{y'(t_0)}{x'(t_0)}$.

Length of a curve

$$L(\gamma) = \int_a^b \|\gamma'(t)\|\,dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\,dt$$

It holds for γ piecewise regular; for a plane curve the term $z'(t)^2$ is absent. If the curve traverses the same arc several times, the formula adds up the lengths of the passages; for simple curves it gives the length of the trace.

Length of a graph and of a polar curve

$$L = \int_a^b \sqrt{1 + f'(x)^2}\,dx, \qquad L = \int_{\theta_1}^{\theta_2} \sqrt{\rho(\theta)^2 + \rho'(\theta)^2}\,d\theta$$

The first formula holds for the graph of f of class C^1 on $[a,b]$. The second holds for the curve $\rho = \rho(\theta)$, that is, $\gamma(\theta) = \big(\rho(\theta)\cos\theta,\ \rho(\theta)\sin\theta\big)$, with ρ of class C^1.

Arc length parameter

$$s(t) = \int_a^t \|\gamma'(\tau)\|\,d\tau, \qquad ds = \|\gamma'(t)\|\,dt$$

For γ regular the function s is strictly increasing from $[a,b]$ onto $[0, L(\gamma)]$. The quantity ds is the length element of line integrals.

Recurring parametrizations

$$\begin{aligned}
\text{segment from } P \text{ to } Q &\qquad \gamma(t) = P + t\,(Q - P), \quad t \in [0,1] \\
\text{circle} &\qquad \gamma(t) = (x_0 + r\cos t,\ y_0 + r\sin t), \quad t \in [0, 2\pi] \\
\text{ellipse} &\qquad \gamma(t) = (a\cos t,\ b\sin t), \quad t \in [0, 2\pi] \\
\text{helix} &\qquad \gamma(t) = (\cos t,\ \sin t,\ t), \quad \|\gamma'(t)\| = \sqrt{2}
\end{aligned}$$

The circle and the ellipse are traversed counterclockwise as t increases; the circle has $\|\gamma'(t)\| = r$.

Line integral of the first kind

$$\int_\gamma f\,ds = \int_a^b f(\gamma(t))\,\|\gamma'(t)\|\,dt$$

It holds for γ regular and f continuous on the trace; for a piecewise regular curve the contributions of the pieces are added. The value depends neither on the parametrization nor on the direction of traversal, and for $f = 1$ it is the length.

Mass and center of mass of a wire

$$m = \int_\gamma \delta\,ds, \qquad \bar{x} = \frac{1}{m}\int_\gamma x\,\delta\,ds, \qquad \bar{y} = \frac{1}{m}\int_\gamma y\,\delta\,ds$$

The linear density $\delta \geq 0$ is continuous on the trace; in space $\bar{z}$ is added. For a homogeneous wire $\bar{x} = \dfrac{1}{L}\displaystyle\int_\gamma x\,ds$; if the wire is homogeneous and the trace is symmetric with respect to a line, the center of mass belongs to that line.

Line integral of the second kind

$$\int_\gamma F \cdot d\gamma = \int_a^b F(\gamma(t)) \cdot \gamma'(t)\,dt = \int_a^b \big(F_1(\gamma(t))\,x'(t) + F_2(\gamma(t))\,y'(t)\big)\,dt$$

It holds for γ regular and $F = (F_1, F_2)$ continuous on the trace; in space $F_3(\gamma(t))\,z'(t)$ is added. With differential forms the integral is written $\displaystyle\int_\gamma F_1\,dx + F_2\,dy$; if F is a force field, the integral is the work.

Line integrals and orientation

$$\int_{-\gamma} f\,ds = \int_\gamma f\,ds, \qquad \int_{-\gamma} F \cdot d\gamma = -\int_\gamma F \cdot d\gamma$$

The opposite curve is $(-\gamma)(t) = \gamma(a + b - t)$ with $t \in [a, b]$. The integral of the second kind therefore depends on the direction of traversal indicated in the exercise.

Conservative fields, potential and Green's theorem

The statements with the proofs are in the chapter Vector fields and Green's theorem.

Conservative field and potential

$$\nabla U = F, \qquad U_x = F_1, \quad U_y = F_2, \quad U_z = F_3$$

The field F is continuous on a connected open set A and the potential U is of class C^1 on A; in the plane the third equality is absent. Two potentials of the same field on A differ by a constant.

Work of a conservative field

$$\int_\gamma F \cdot d\gamma = U(Q) - U(P), \qquad \oint_\gamma F \cdot d\gamma = 0 \ \text{ if } \gamma \text{ is closed}$$

It holds for $F = \nabla U$ on A and γ piecewise regular in A from P to Q. Therefore, if the circulation of F along a closed curve contained in A is different from zero, F is not conservative on A.

Irrotational field

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \ \text{ in } A, \qquad \operatorname{curl} F = (0,0,0) \ \text{ in } A \subseteq \mathbb{R}^3$$

For F of class C^1 the condition is *necessary* for F to be conservative. If the cross derivatives are different, the field is therefore not conservative.

Criterion for conservative fields

$$\begin{cases} F \text{ of class } C^1 \text{ and irrotational in } A \\ A \text{ simply connected} \end{cases} \implies F \text{ conservative in } A$$

Simply connected sets are $\mathbb{R}^2$, $\mathbb{R}^3$, convex sets, half-planes and $\mathbb{R}^3$ with a point removed. Not simply connected are $\mathbb{R}^2 \setminus \{(0,0)\}$, annuli and $\mathbb{R}^3$ with a line removed.

Potential by successive integration

$$U(x,y) = \int F_1(x,y)\,dx + c(y), \qquad c'(y) = F_2(x,y) - \frac{\partial}{\partial y}\int F_1(x,y)\,dx$$

The antiderivative in x is computed with y fixed. If the field is conservative, the right-hand side of the equation for $c'(y)$ depends only on y; integrating in y gives $c(y)$ up to a constant.

Potential as work along a polygonal path

$$U(x,y) = \int_{x_0}^{x} F_1(t, y_0)\,dt + \int_{y_0}^{y} F_2(x,t)\,dt$$

$$U(x,y,z) = \int_{x_0}^{x} F_1(t, y_0, z_0)\,dt + \int_{y_0}^{y} F_2(x,t,z_0)\,dt + \int_{z_0}^{z} F_3(x,y,t)\,dt$$

They hold for F conservative on an open set containing the polygonal path from (x_0, y_0) to (x, y) with sides parallel to the axes, in space from (x_0, y_0, z_0) to (x, y, z). Often the starting point is the origin.

Green's theorem

$$\oint_{\partial D} F_1\,dx + F_2\,dy = \iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) dx\,dy$$

It holds for D a regular domain with boundary ∂D piecewise regular and positively oriented, and for $F = (F_1, F_2)$ of class C^1 on an open set containing D. The boundary is positively oriented if it leaves D on its left: counterclockwise on the outer boundary, clockwise on the boundaries of the holes.

Area with Green's formula

$$\text{area}(D) = \oint_{\partial D} x\,dy = -\oint_{\partial D} y\,dx = \frac{1}{2}\oint_{\partial D} (x\,dy - y\,dx)$$

They hold for D a regular domain with positively oriented boundary. For the ellipse $r(t) = (a\cos t,\ b\sin t)$, $t \in [0, 2\pi]$, the third formula gives πab.

Warning

An irrotational field on an open set that is not simply connected may fail to be conservative. The field $F(x,y) = \left(\dfrac{-y}{x^2+y^2},\ \dfrac{x}{x^2+y^2}\right)$ is irrotational on $\mathbb{R}^2 \setminus \{(0,0)\}$, but its circulation along the unit circle traversed counterclockwise equals 2π. Therefore the field is not conservative on $\mathbb{R}^2 \setminus \{(0,0)\}$; on the half-plane $x > 0$ a potential is $\arctan \dfrac{y}{x}$.

Double and triple integrals and reduction formulas

The proofs and the examples are in the chapter Double and triple integrals.

Fubini's theorem on rectangles

$$\iint_{[a,b]\times[c,d]} f(x,y)\,dx\,dy = \int_a^b \left(\int_c^d f(x,y)\,dy\right) dx = \int_c^d \left(\int_a^b f(x,y)\,dx\right) dy$$

It holds for f continuous on the rectangle. The two orders of integration give the same value.

Functions with separated variables

$$\iint_{[a,b]\times[c,d]} g(x)\,h(y)\,dx\,dy = \left(\int_a^b g(x)\,dx\right)\left(\int_c^d h(y)\,dy\right)$$

It holds for g continuous on $[a,b]$ and h continuous on $[c,d]$, *only* on rectangular domains. The same holds for three factors on a rectangular box.

Reduction formulas on normal domains

$$D = \left\{(x,y) : \begin{cases} a \le x \le b \\ g_1(x) \le y \le g_2(x) \end{cases}\right\} \implies \iint_D f\,dx\,dy = \int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right) dx$$

$$D = \left\{(x,y) : \begin{cases} c \le y \le d \\ h_1(y) \le x \le h_2(y) \end{cases}\right\} \implies \iint_D f\,dx\,dy = \int_c^d \left(\int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\right) dy$$

They hold for f continuous on D and continuous functions $g_1 \le g_2$, $h_1 \le h_2$. The outer variable ranges over a fixed interval.

Changing the order of integration

$$\int_0^1 \left(\int_x^1 f(x,y)\,dy\right) dx = \int_0^1 \left(\int_0^y f(x,y)\,dx\right) dy$$

The two formulations describe the same triangle with vertices $(0,0)$, $(0,1)$, $(1,1)$, normal with respect to both axes. In the general case one draws the domain and describes it with respect to the other axis.

Linearity, additivity and monotonicity

$$\iint_{D_1\cup D_2} f = \iint_{D_1} f + \iint_{D_2} f, \qquad \left|\iint_D f\right| \le \iint_D |f|$$

Additivity holds if D_1 and D_2 have no common interior points. The integral is linear, and if $f \le g$ on D then $\iint_D f \le \iint_D g$.

Symmetries of the domain

$$f(-x,y) = -f(x,y) \implies \iint_D f\,dx\,dy = 0, \qquad f(-x,y) = f(x,y) \implies \iint_D f\,dx\,dy = 2\iint_{D^+} f$$

They hold for f continuous on D symmetric with respect to the y-axis, with D^+ the part of D where $x \geq 0$. Exchanging x and y gives the formulas for domains symmetric with respect to the x-axis; in triple integrals the analogous formulas hold with respect to the coordinate planes.

Integration by vertical lines

$$\iiint_E f\,dx\,dy\,dz = \iint_D \left(\int_{\alpha(x,y)}^{\beta(x,y)} f(x,y,z)\,dz \right) dx\,dy$$

It holds for $E = \{(x,y,z) : (x,y) \in D,\ \alpha(x,y) \leq z \leq \beta(x,y)\}$, with $\alpha \leq \beta$ continuous on D and f continuous on E. It is used for solids lying between two graphs.

Integration by slices

$$\iiint_E f\,dx\,dy\,dz = \int_a^b \left(\iint_{E_z} f(x,y,z)\,dx\,dy \right) dz, \qquad \iiint_E g(z)\,dx\,dy\,dz = \int_a^b g(z)\ \text{Area}(E_z)\,dz$$

They hold for E lying between the planes $z = a$ and $z = b$, with E_z the cross section at height z and the integrands continuous. The second formula is used when the function depends only on z.

Changes of variables and the Jacobian

The formulas are explained, with the proofs and the examples, in the chapter Double and triple integrals.

Change of variables

$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f\big(x(u,v),\, y(u,v)\big)\,|J(u,v)|\,du\,dv, \qquad J = \det \begin{pmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{pmatrix}$$

It holds for $\Phi(u,v) = \big(x(u,v), y(u,v)\big)$ of class C^1, injective in the interior of D', with $J \neq 0$ in the interior of D', $D = \Phi(D')$ and f continuous on D. In triple integrals J is the determinant of the 3×3 matrix of the partial derivatives.

Jacobian of the inverse transformation

$$\det \frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{\det \dfrac{\partial(u,v)}{\partial(x,y)}}$$

It holds when the new variables are assigned as $u = u(x,y)$, $v = v(x,y)$ and the determinant in the denominator is nonzero. For $u = xy$ and $v = \dfrac{y}{x}$ in the first quadrant one obtains $dx\,dy = \dfrac{du\,dv}{2v}$.

Linear transformations

$$u = a_1 x + b_1 y, \quad v = a_2 x + b_2 y \implies dx\,dy = \frac{du\,dv}{|a_1 b_2 - a_2 b_1|}$$

It holds for $a_1 b_2 - a_2 b_1 \neq 0$. A parallelogram with sides on the lines along which u or v is constant becomes a rectangle in the uv-plane.

Polar coordinates

$$x = \rho\cos\theta, \qquad y = \rho\sin\theta, \qquad dx\,dy = \rho\,d\rho\,d\theta$$

They hold for $\rho \geq 0$ and $\theta \in [0, 2\pi)$, with $x^2 + y^2 = \rho^2$. The disk centered at the origin with radius R becomes $0 \leq \rho \leq R$, $0 \leq \theta \leq 2\pi$; the annulus with radii $R_1 < R_2$ becomes $R_1 \leq \rho \leq R_2$. Centered at (x_0, y_0) they are written $x = x_0 + \rho\cos\theta$, $y = y_0 + \rho\sin\theta$, with the same Jacobian.

Elliptic coordinates

$$x = a\rho\cos\theta, \qquad y = b\rho\sin\theta, \qquad dx\,dy = ab\rho\,d\rho\,d\theta$$

They hold for $a, b > 0$, $\rho \geq 0$ and $\theta \in [0, 2\pi)$. The ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} \leq 1$ becomes $0 \leq \rho \leq 1$, $0 \leq \theta \leq 2\pi$, and its area is πab.

Cylindrical coordinates

$$x = \rho\cos\theta, \qquad y = \rho\sin\theta, \qquad z = z, \qquad dx\,dy\,dz = \rho\,d\rho\,d\theta\,dz$$

They hold for $\rho \geq 0$ and $\theta \in [0, 2\pi)$. They are used for solids with the z-axis as axis of symmetry: the cylinder $x^2 + y^2 \leq R^2$, $0 \leq z \leq h$ becomes a rectangular box and the cone $z = \sqrt{x^2 + y^2}$ has equation $z = \rho$.

Spherical coordinates

$$x = r\sin\varphi\cos\theta, \quad y = r\sin\varphi\sin\theta, \quad z = r\cos\varphi, \qquad dx\,dy\,dz = r^2\sin\varphi\,dr\,d\varphi\,d\theta$$

They hold for $r \geq 0$, $\varphi \in [0, \pi]$ the colatitude measured from the positive z half-axis and $\theta \in [0, 2\pi)$, with $x^2 + y^2 + z^2 = r^2$. The ball of radius R becomes $0 \leq r \leq R$, $0 \leq \varphi \leq \pi$, $0 \leq \theta \leq 2\pi$; the half ball with $z \geq 0$ has $0 \leq \varphi \leq \dfrac{\pi}{2}$.

Warning

The area element in polar coordinates contains the factor ρ of the Jacobian. For the disk of radius R one has $\displaystyle\int_0^{2\pi}\int_0^{R} \rho\,d\rho\,d\theta = \pi R^2$; without the factor ρ one would obtain $2\pi R$, which is the length of the circle.

Areas, volumes, masses and centers of mass

The definitions and computation examples are in the chapter Double and triple integrals.

Area and volume as integrals

$$\text{Area}(D) = \iint_D dx\,dy, \qquad \text{Vol}(E) = \iiint_E dx\,dy\,dz, \qquad \text{Vol} = \iint_D \big(f(x,y) - g(x,y)\big)\,dx\,dy$$

The third formula gives the volume of the solid lying between the graphs of g and f over D, with $g \leq f$ continuous. For $g = 0$ and $f \geq 0$ it is the volume under the graph of f.

Mass and center of mass of a lamina

$$m = \iint_D \delta\,dx\,dy, \qquad \bar{x} = \frac{1}{m}\iint_D x\,\delta\,dx\,dy, \qquad \bar{y} = \frac{1}{m}\iint_D y\,\delta\,dx\,dy$$

The surface density $\delta(x, y) \geq 0$ is continuous on D. If the lamina is homogeneous, δ cancels and $\bar{x} = \dfrac{1}{\text{Area}(D)}\displaystyle\iint_D x\,dx\,dy$; if the lamina is homogeneous and D is symmetric with respect to a line, the center of mass belongs to that line.

Mass and center of mass of a solid

$$m = \iiint_E \delta\,dx\,dy\,dz, \qquad \bar{z} = \frac{1}{m}\iiint_E z\,\delta\,dx\,dy\,dz$$

The density $\delta(x, y, z) \geq 0$ is continuous on E; the coordinates $\bar{x}$ and $\bar{y}$ are written in the same way with x and y in place of z.

Standard volumes

Solid	Volume
ball of radius R	$\frac{4}{3}\pi R^3$
cylinder of radius R and height h	$\pi R^2 h$
cone of radius R and height h	$\frac{1}{3}\pi R^2 h$
ellipsoid with semi-axes a, b, c	$\frac{4}{3}\pi abc$
$x^2 + y^2 \le z \le h$	$\frac{1}{2}\pi h^2$

The values are obtained with spherical and cylindrical coordinates, with integration by slices and, for the ellipsoid, with the transformation $x = au$, $y = bv$, $z = cw$, whose Jacobian is abc.

Parametric surfaces and surface integrals

The proofs are in the chapter Surfaces, flux and the theorems of Gauss and Stokes.

Normal vector and regular surface

$$\mathbf{N}(u,v) = \mathbf{r}_u(u,v) \wedge \mathbf{r}_v(u,v), \qquad \mathbf{n} = \frac{\mathbf{N}}{\|\mathbf{N}\|}$$

They hold for $\mathbf{r} : D \to \mathbb{R}^3$ of class C^1. The surface is regular if $\mathbf{r}$ is injective at the interior points of D and $\mathbf{N} \neq \mathbf{0}$ at the interior points of D.

Cross product

$$\mathbf{r}_u \wedge \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_u & y_u & z_u \\ x_v & y_v & z_v \end{vmatrix} = \left(y_u z_v - z_u y_v,\ \ z_u x_v - x_u z_v,\ \ x_u y_v - y_u x_v\right)$$

The resulting vector is orthogonal to $\mathbf{r}_u$ and to $\mathbf{r}_v$, and its norm is the area of the parallelogram having them as sides.

Tangent plane to a parametric surface

$$N_1\,(x - x_0) + N_2\,(y - y_0) + N_3\,(z - z_0) = 0$$

It holds at the point $P_0 = \mathbf{r}(u_0, v_0) = (x_0, y_0, z_0)$, with (u_0, v_0) interior to D and $\mathbf{N}(u_0, v_0) = (N_1, N_2, N_3) \neq \mathbf{0}$.

Cartesian surface

$$\mathbf{r}(x,y) = \big(x,\, y,\, f(x,y)\big), \qquad \mathbf{N} = \big(-f_x,\, -f_y,\, 1\big), \qquad \|\mathbf{N}\| = \sqrt{1 + f_x^2 + f_y^2}$$

It holds for f of class C^1 on D. The surface is always regular and $\mathbf{N}$ points upward.

Parametrizations of the sphere and the cylinder

$$\mathbf{r}(\varphi,\theta) = \big(R\sin\varphi\cos\theta,\; R\sin\varphi\sin\theta,\; R\cos\varphi\big), \qquad \|\mathbf{N}\| = R^2\sin\varphi$$

$$\mathbf{r}(\theta,t) = \big(R\cos\theta,\; R\sin\theta,\; t\big), \qquad \mathbf{N} = (R\cos\theta,\; R\sin\theta,\; 0)$$

For the sphere of radius R one has $(\varphi,\theta) \in [0,\pi]\times[0,2\pi]$ and $\mathbf{N} = \mathbf{r}_\varphi \wedge \mathbf{r}_\theta = R\sin\varphi\,\mathbf{r}(\varphi,\theta)$, which points outward. For the cylinder $x^2+y^2=R^2$ one has $(\theta,t)\in[0,2\pi]\times[0,h]$ and $\mathbf{N} = \mathbf{r}_\theta \wedge \mathbf{r}_t$ points outward.

Area of a surface

$$\mathrm{Area}(\Sigma) = \iint_D \|\mathbf{r}_u \wedge \mathbf{r}_v\|\, du\, dv, \qquad d\sigma = \|\mathbf{r}_u \wedge \mathbf{r}_v\|\, du\, dv$$

It holds for $\Sigma = \mathbf{r}(D)$ regular; the value does not depend on the parametrization. The quantity $d\sigma$ is the area element.

Area of a Cartesian surface

$$\mathrm{Area}(\Sigma) = \iint_D \sqrt{1 + f_x^2 + f_y^2}\, dx\, dy$$

It holds for the graph of f of class C^1 on D. For the paraboloid $z = x^2+y^2$ over the unit disk one obtains $\dfrac{\pi}{6}\big(5\sqrt{5}-1\big)$.

Area of a surface of revolution

$$\mathrm{Area}(\Sigma) = 2\pi \int_a^b g(t)\,\sqrt{g'(t)^2 + h'(t)^2}\, dt$$

Σ is obtained by rotating through one full turn about the z-axis the regular curve $x = g(t)$, $z = h(t)$, $t\in[a,b]$, of the half-plane $y=0$, with $g \geq 0$ and g, h of class C^1.

Standard areas

Surface	Area
sphere of radius R	$4\pi R^2$
spherical cap of height h	$2\pi Rh$
lateral surface of the cylinder	$2\pi Rh$
lateral surface of the cone	$\pi R\sqrt{R^2+h^2}$

R is the radius of the sphere, of the cylinder and of the base of the cone, h the height of the cap, of the cylinder and of the cone.

Surface integral

$$\iint_\Sigma f\,d\sigma = \iint_D f\big(\mathbf{r}(u,v)\big)\,\|\mathbf{r}_u \wedge \mathbf{r}_v\|\,du\,dv$$

It holds for $\Sigma = \mathbf{r}(D)$ regular and f continuous on Σ. For $f = 1$ one obtains the area; if f is a surface density, the integral is the mass of the surface.

Flux, divergence and curl, theorems of Gauss and Stokes

The statements and the proofs of the theorems are in the chapter Surfaces, flux and the theorems of Gauss and Stokes.

Flux of a vector field

$$\Phi_\Sigma(\mathbf{F}) = \iint_\Sigma \mathbf{F}\cdot\mathbf{n}\,d\sigma = \iint_D \mathbf{F}\big(\mathbf{r}(u,v)\big)\cdot\big(\mathbf{r}_u \wedge \mathbf{r}_v\big)\,du\,dv$$

It holds for Σ regular, oriented by $\mathbf{n}$, and $\mathbf{F}$ continuous on an open set containing Σ. The second equality holds if $\mathbf{r}_u \wedge \mathbf{r}_v$ has the direction of $\mathbf{n}$; with the opposite direction the flux changes sign.

Flux through a Cartesian surface

$$\Phi_\Sigma(\mathbf{F}) = \iint_D \big(-F_1\,f_x - F_2\,f_y + F_3\big)\,dx\,dy$$

It holds for the graph of f of class C^1 on D oriented upward, with the components of $\mathbf{F}$ evaluated at $(x, y, f(x,y))$. With the downward orientation the flux changes sign.

Divergence

$$\operatorname{div}\mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

It is defined for $\mathbf{F}$ of class C^1. The field is solenoidal if $\operatorname{div}\mathbf{F} = 0$; for $\mathbf{F} = (x, y, z)$ one has $\operatorname{div}\mathbf{F} = 3$.

Curl

$$\operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$$

It is defined for $\mathbf{F}$ of class C^1. For a plane field $(F_1, F_2, 0)$ with components independent of z only the third component remains.

Identities for divergence and curl

$$\operatorname{curl}(\nabla f) = \mathbf{0}, \qquad \operatorname{div}(\operatorname{curl}\mathbf{F}) = 0$$

They hold for f and $\mathbf{F}$ of class C^2 on an open subset of $\mathbb{R}^3$. Every gradient field is therefore irrotational, and every curl is solenoidal.

Divergence theorem (Gauss)

$$\iint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_V \operatorname{div}\mathbf{F} \, dx\, dy\, dz$$

It holds for V a regular domain in space with boundary oriented by the **outward** unit normal and $\mathbf{F}$ of class C^1 on an open set containing V. For $\mathbf{F} = (x, y, z)$ the outward flux is $3\,\mathrm{Vol}(V)$.

Flux through an open surface

$$\Phi_{\Sigma}(\mathbf{F}) = \iiint_V \operatorname{div}\mathbf{F} \, dx\, dy\, dz - \Phi_{\Sigma'}(\mathbf{F})$$

It holds when the auxiliary surface Σ' closes Σ, that is, $\Sigma \cup \Sigma'$ is the boundary of the regular domain V, with both surfaces oriented by the outward normal to V. Usually Σ' is a flat disk.

Stokes' theorem

$$\oint_{\partial\Sigma} \mathbf{F} \cdot d\mathbf{r} = \iint_{\Sigma} \operatorname{curl} \mathbf{F} \cdot \mathbf{n} \, d\sigma$$

It holds for Σ regular and oriented, with boundary $\partial\Sigma$ closed, piecewise regular and coherently oriented, and for $\mathbf{F}$ of class C^1 on an open set containing Σ. The orientation is coherent if an observer standing as $\mathbf{n}$ who walks along the boundary has the surface on his left; for a graph oriented upward the boundary is traversed counterclockwise when seen from above.

Sequences and series of functions

The proofs and the counterexamples are in the chapter Sequences and series of functions.

Pointwise convergence

$$\lim_{n \to +\infty} f_n(x) = f(x) \qquad \text{for every } x \in A$$

The limit is computed for fixed x, with the methods for limits of sequences, distinguishing cases when the expression requires it. The function f is the candidate limit for uniform convergence.

Uniform convergence and the sup criterion

$$f_n \to f \text{ uniformly on } A \iff M_n = \sup_{x \in A} |f_n(x) - f(x)| \xrightarrow[n \to +\infty]{} 0$$

The value M_n is usually found by studying the derivative of $f_n - f$ on A. Uniform convergence implies pointwise convergence, and the answer depends on the set A.

Continuity of the uniform limit

$$f_n \text{ continuous on } A,\ f_n \to f \text{ uniformly on } A \implies f \text{ continuous on } A$$

Therefore, if the pointwise limit of continuous functions is discontinuous, the convergence is not uniform.

Passing to the limit under the integral sign

$$\lim_{n \to +\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx, \qquad \left| \int_a^b f_n - \int_a^b f \right| \le (b-a)\, M_n$$

It holds for f_n continuous on $[a, b]$ with $f_n \to f$ uniformly on $[a, b]$.

Differentiation of sequences

$$\begin{cases} f_n(x_0) \text{ converges for at least one } x_0 \in [a,b] \\ f_n' \to g \text{ uniformly on } [a,b] \end{cases} \implies \left(\lim_{n\to+\infty} f_n \right)' = \lim_{n\to+\infty} f_n'$$

It holds for $f_n \in C^1([a,b])$. Under the hypotheses f_n converges uniformly on $[a,b]$ to a function f of class C^1 with $f' = g$.

Series of functions: partial sums and remainder

$$s_N(x) = \sum_{n=1}^{N} f_n(x), \qquad \sup_{x\in A} \left| \sum_{n=N+1}^{\infty} f_n(x) \right| \xrightarrow[N\to+\infty]{} 0$$

The series converges pointwise at x if the numerical series $\sum f_n(x)$ converges. The condition on the remainder is equivalent to uniform convergence on A.

Total convergence (Weierstrass M-test)

$$\begin{cases} |f_n(x)| \le M_n \ \text{ for every } x \in A \\ \sum_{n=1}^{\infty} M_n \ \text{ converges} \end{cases} \implies \sum_{n=1}^{\infty} f_n \text{ converges totally, hence uniformly, on } A$$

The sharpest choice is $M_n = \sup_A |f_n|$. The chain total $\implies$ uniform $\implies$ pointwise holds, and total convergence also implies absolute convergence at every point of A.

Numerical series for comparison

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges } \iff p > 1, \qquad \sum_{n=0}^{\infty} q^n = \frac{1}{1-q} \ \text{ for } |q| < 1$$

These are the series used most often for the sequence M_n of total convergence.

Remainder of an alternating series

$$\left| \sum_{n=1}^{\infty} (-1)^n a_n - \sum_{n=1}^{N} (-1)^n a_n \right| \le a_{N+1}$$

It holds for (a_n) decreasing with $a_n \to 0$. If $a_n = a_n(x)$ and $\sup_A a_{N+1}(x) \to 0$, the series converges uniformly on A even when the convergence is not total.

Continuity, term-by-term integration and differentiation

$$\int_a^b \sum_{n=1}^{\infty} f_n(x)\,dx = \sum_{n=1}^{\infty} \int_a^b f_n(x)\,dx, \qquad \left(\sum_{n=1}^{\infty} f_n\right)' = \sum_{n=1}^{\infty} f_n'$$

If the f_n are continuous and the series converges uniformly on $[a,b]$, the sum is continuous and the first equality holds. The second holds for $f_n \in C^1([a,b])$ if $\sum f_n(x_0)$ converges at at least one point and $\sum f_n'$ converges uniformly on $[a,b]$, usually totally.

Warning

Pointwise convergence does not preserve continuity. The functions $f_n(x) = x^n$ are continuous on $[0,1]$ and converge pointwise to the function equal to 0 on $[0,1)$ and to 1 at $x = 1$, which is discontinuous. On $[0,1)$ one has $\sup x^n = 1$ for every n, hence the convergence is not uniform; on $[0,a]$ with $0 < a < 1$ one has $M_n = a^n \to 0$ and the convergence is uniform.

Power series and radius of convergence

The proofs of the theorems on the radius are in the chapter Power series and Taylor series.

Radius and set of convergence

$$|x - x_0| < R \implies \sum_{n=0}^{\infty} a_n (x - x_0)^n \text{ converges absolutely}$$

$$|x - x_0| > R \implies \sum_{n=0}^{\infty} a_n (x - x_0)^n \text{ does not converge}$$

$$(x_0 - R,\ x_0 + R) \subseteq E \subseteq [x_0 - R,\ x_0 + R]$$

The radius $R \in [0, +\infty]$ exists for every power series; the inclusions for the set of convergence E hold for $0 < R < +\infty$. At the points $x_0 \pm R$ the theorem gives no information and the two numerical series are studied separately.

Radius with the ratio test

$$\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = \ell \implies R = \frac{1}{\ell}$$

It holds for $a_n \neq 0$ eventually and $\ell \in [0, +\infty]$, with $R = +\infty$ if $\ell = 0$ and $R = 0$ if $\ell = +\infty$. For $\sum \dfrac{x^n}{n!}$ one has $\ell = 0$, hence $R = +\infty$.

Radius with the root test

$$\lim_{n\to\infty}\sqrt[n]{|a_n|}=\ell \implies R=\frac{1}{\ell}$$

It holds with the same conventions as the ratio test. It is used when a_n is an n-th power: for $a_n=\left(\dfrac{n}{2n+1}\right)^n$ one has $\ell=\dfrac{1}{2}$, hence $R=2$.

Lacunary series

$$\sum_{n=0}^{\infty}a_n x^{kn}=\sum_{n=0}^{\infty}a_n t^n, \qquad t=x^k, \qquad R=\sqrt[k]{R_t}$$

R is the radius of the series in x, R_t that of the series in t and $k\geq 2$ is an integer. The ratio and root formulas on the coefficients do not apply to the series in x, which has infinitely many zero coefficients. For $\sum\dfrac{x^{2n}}{4^n}$ one has $R_t=4$, hence $R=2$.

Behavior at the endpoints

$$\sum_{n=1}^{\infty}x^n: \quad E=(-1,1)$$
$$\sum_{n=1}^{\infty}\frac{x^n}{n}: \quad E=[-1,1)$$
$$\sum_{n=1}^{\infty}\frac{x^n}{n^2}: \quad E=[-1,1]$$

The three series have radius 1. At $x=x_0\pm R$ the numerical series is studied with the comparison test, the limit comparison test and the Leibniz test.

Properties of the sum

$$f(x)=\sum_{n=0}^{\infty}a_n(x-x_0)^n \text{ is continuous on } (x_0-R,\ x_0+R)$$

It holds for $R>0$, because for *every* $0<r<R$ the series converges totally on $[x_0-r,x_0+r]$. If $R<+\infty$ and the series converges at x_0+R, by Abel's theorem $\lim\limits_{x\to(x_0+R)^-}f(x)=\sum\limits_{n=0}^{\infty}a_nR^n$; the same holds at x_0-R.

Term-by-term differentiation and integration

$$f'(x) = \sum_{n=1}^{\infty} n\, a_n (x - x_0)^{n-1}, \qquad \int_{x_0}^{x} f(t)\, dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} (x - x_0)^{n+1}$$

They hold for every $x \in (x_0 - R,\, x_0 + R)$ with $R > 0$. The two series have the same radius R as the original series, and f is of class C^∞ on the open interval.

Taylor series expansions

Taylor series and their validity are treated in the chapter Power series and Taylor series.

Taylor series and uniqueness of the expansion

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n, \qquad f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n \implies a_n = \frac{f^{(n)}(x_0)}{n!}$$

The series is defined for f of class C^∞ near x_0; for $x_0 = 0$ it is called the Maclaurin series. The implication holds if the expansion is valid on an interval $(x_0 - \delta, x_0 + \delta)$ with $\delta > 0$; therefore every expansion obtained from known expansions is the Taylor series of the function.

Criterion for expandability

$$|f^{(n)}(x)| \le M L^n \text{ for every } n \text{ and every } x \in I \implies f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \text{ on } I$$

It holds for f of class C^∞ on the interval I centered at x_0 and constants $M, L > 0$. The expansion holds *if and only if* the remainder $f(x) - T_n(x)$ tends to 0 as $n \to \infty$.

Exponential, sine and cosine

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n\, x^{2n}}{(2n)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$$

The three expansions hold for every $x \in \mathbb{R}$.

Geometric series and related expansions

$$\frac{1}{1-x}=\sum_{n=0}^{\infty}x^n, \qquad \frac{1}{1+x}=\sum_{n=0}^{\infty}(-1)^n x^n, \qquad \frac{1}{1+x^2}=\sum_{n=0}^{\infty}(-1)^n x^{2n}$$

The three expansions hold for $-1<x<1$.

Logarithm and arctangent

$$\begin{aligned}
\ln(1+x)&=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}x^n}{n} && -1<x\le 1\\
-\ln(1-x)&=\sum_{n=1}^{\infty}\frac{x^n}{n} && -1\le x<1\\
\arctan x&=\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1} && -1\le x\le 1\\
\ln\frac{1+x}{1-x}&=2\sum_{n=0}^{\infty}\frac{x^{2n+1}}{2n+1} && -1<x<1
\end{aligned}$$

The included endpoints follow from Abel's theorem; in particular $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}=\ln 2$ and $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}=\frac{\pi}{4}$.

Binomial series

$$(1+x)^{\alpha}=\sum_{n=0}^{\infty}\binom{\alpha}{n}x^n, \qquad \binom{\alpha}{n}=\frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}, \qquad \binom{\alpha}{0}=1$$

$$\sqrt{1+x}=1+\frac{x}{2}-\frac{x^2}{8}+\frac{x^3}{16}-\cdots, \qquad \frac{1}{\sqrt{1+x}}=1-\frac{x}{2}+\frac{3x^2}{8}-\frac{5x^3}{16}+\cdots$$

The expansion holds for $-1<x<1$ and $\alpha\in\mathbb{R}$. For $\alpha\in\mathbb{N}$ the sum is finite and coincides with the binomial theorem, valid for every x.

Expansions by substitution

$$g(t)=\sum_{n=0}^{\infty}a_n t^n \ \text{ for } |t|<R \implies g(c\,x^k)=\sum_{n=0}^{\infty}a_n c^n x^{kn} \ \text{ for } |c\,x^k|<R$$

It holds for $c\neq 0$ and $k\ge 1$ an integer; the region of validity is obtained by imposing on the substituted expression the condition of the original expansion. For example $e^{-x^2}=\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{n!}$ for every $x\in\mathbb{R}$.

Expansions centered at a point other than the origin

$$e^x = \sum_{n=0}^{\infty} \frac{e^{x_0}}{n!}(x-x_0)^n \qquad x \in \mathbb{R}$$

$$\frac{1}{x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{x_0^{n+1}}(x-x_0)^n \qquad |x-x_0| < x_0$$

$$\ln x = \ln x_0 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\,x_0^n}(x-x_0)^n \qquad 0 < x \le 2x_0$$

The last two hold for $x_0 > 0$. They are obtained from the expansions at 0 with the substitution $t = \dfrac{x-x_0}{x_0}$, or with $t = x - x_0$ for the exponential.

Sums obtained by differentiation

$$\sum_{n=1}^{\infty} n\,x^{n-1} = \frac{1}{(1-x)^2}, \qquad \sum_{n=1}^{\infty} n\,x^n = \frac{x}{(1-x)^2}, \qquad \sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3}$$

They hold for $-1 < x < 1$. A numerical series with terms of this form is summed by substituting the value of x: for $x = \dfrac{1}{2}$ one obtains $\displaystyle\sum_{n=1}^{\infty} \frac{n}{2^n} = 2$ and $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{2^n} = 6$.

Fourier series

The proofs and the examples are in the chapter Fourier series.

Fourier coefficients and Fourier series

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \quad (n \ge 0), \qquad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)\,dx \quad (n \ge 1)$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\big(a_n\cos(nx) + b_n\sin(nx)\big)$$

They hold for f of period 2π, integrable on $[-\pi, \pi]$. The term $\dfrac{a_0}{2}$ is the mean of f over a period; the integral can be computed over any interval of length 2π.

Even functions and odd functions

$$f \text{ even} \implies b_n = 0, \quad a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,dx$$

$$f \text{ odd} \implies a_n = 0, \quad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,dx$$

For an even function the series contains only cosines, for an odd function only sines. If f is modified at finitely many points, the coefficients remain the same.

Orthogonality of the trigonometric system

$$\int_{-\pi}^{\pi} \cos(nx)\cos(mx)\,dx = \int_{-\pi}^{\pi} \sin(nx)\sin(mx)\,dx = 0 \quad (n \neq m)$$

$$\int_{-\pi}^{\pi} \cos^2(nx)\,dx = \int_{-\pi}^{\pi} \sin^2(nx)\,dx = \pi, \qquad \int_{-\pi}^{\pi} \sin(nx)\cos(mx)\,dx = 0$$

They hold for every pair of integers $n, m \geq 1$; moreover $\int_{-\pi}^{\pi} \cos(nx)\,dx = \int_{-\pi}^{\pi} \sin(nx)\,dx = 0$.

Recurring integrals in the coefficients

$$\int x\cos(nx)\,dx = \frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2} + c$$

$$\int x\sin(nx)\,dx = -\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} + c$$

$$\cos(n\pi) = (-1)^n, \qquad \sin(n\pi) = 0$$

The antiderivatives hold for $n \geq 1$ an integer and are obtained by integration by parts; the values at $n\pi$ hold for every integer n.

Pointwise convergence

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n\cos(nx) + b_n\sin(nx)\big) = \frac{f(x^+) + f(x^-)}{2}$$

It holds at every $x \in \mathbb{R}$ for f of period 2π and piecewise smooth. At the points of continuity the sum is $f(x)$; at the jump points it is the average of the one-sided limits, and for the extension of a function given on $[-\pi, \pi)$ the jumps may be located at $\pm\pi$.

Uniform convergence

$$\sum_{n=1}^{\infty} \big(|a_n| + |b_n|\big) \text{ converges}, \qquad |a_n\cos(nx) + b_n\sin(nx)| \leq |a_n| + |b_n|$$

They hold for f of period 2π, **continuous on** $\mathbb{R}$ and piecewise smooth. By the Weierstrass M-test the series then converges totally, hence uniformly, to f on $\mathbb{R}$. For a function given on $[-\pi, \pi]$ the continuity of the extension requires $f(-\pi) = f(\pi)$.

Parseval's identity

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right)$$

It holds for f of period 2π with square integrable on $[-\pi,\pi]$. It is used to sum numerical series with the squares of the coefficients.

Period 2L

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx, \qquad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\left(\frac{n\pi x}{L}\right) + b_n\sin\left(\frac{n\pi x}{L}\right)\right)$$

They hold for f of period $2L$, with $L>0$, integrable on $[-L,L]$. The convergence theorems and the formulas for even and odd functions remain valid; in Parseval's identity the left-hand side becomes $\frac{1}{L}\int_{-L}^{L} f(x)^2\,dx$.

Standard Fourier expansions

$$x \sim \sum_{n=1}^{\infty}\frac{2\,(-1)^{n+1}}{n}\sin(nx) \qquad x\in[-\pi,\pi)$$

$$x^2 \sim \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\frac{4\,(-1)^n}{n^2}\cos(nx) \qquad x\in[-\pi,\pi]$$

$$|x| \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\cos\big((2k+1)x\big)}{(2k+1)^2} \qquad x\in[-\pi,\pi]$$

$$q(x) \sim \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\sin\big((2k+1)x\big)}{2k+1} \qquad x\in[-\pi,\pi)$$

The expansions refer to the 2π-periodic extensions of the functions indicated. In the last line q is the square wave, equal to -1 on $[-\pi,0)$ and to 1 on $[0,\pi)$; its coefficients are $b_n = \dfrac{2\big(1-(-1)^n\big)}{n\pi}$.

Sums of numerical series

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}, \qquad \sum_{k=0}^{\infty}\frac{1}{(2k+1)^2} = \frac{\pi^2}{8}, \qquad \sum_{n=1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}$$

The first three are obtained from the pointwise convergence of the expansions of x^2 at $x=\pi$ and at $x=0$ and of $|x|$ at $x=0$; the fourth from Parseval's identity for x^2.

Warning

If f has a jump point, the Fourier series does not converge uniformly on $\mathbb{R}$, because the partial sums are continuous and the uniform limit of continuous functions is continuous. For the extension of $f(x) = x$ from $[-\pi, \pi)$ the one-sided limits at $x = \pi$ are π and $-\pi$, and the series at $x = \pi$ converges to the average 0; the convergence is therefore only pointwise.